



WOLLO University

Department of software engineering

Blueprint based exit material for

Fundamental networking

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❖ Network protocol and standard

A protocol is a set of rules that governs data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated. For instance, for one computer to send a message to another computer, the first computer must perform the following general steps.

- break the data into small sections called packets
- add addressing information to the packets identifying the destination computer
- deliver the data to the network card for transmission over the network

The receiving computer must perform the same steps, but in reverse order

- accept the data from the NIC
- remove transmitting information that was added by the transmitting computer
- reassemble the packets of data in to the original message

The key elements of a protocol are: syntax: refers to the structure or format of the data, semantics: refers to the meaning of each section of bits and timing: refers to when data should be sent and how fast they can be sent.

A network protocol is a set of rules that define how data is transmitted between different devices in a network. It ensures that the devices can communicate with each other, regardless of their internal processes, structure, or design.

Network standards are documents that define the specific implementation of a protocol. They are developed by organizations such as the Internet Engineering Task Force (IETF) and the Institute of Electrical and Electronics Engineers (IEEE).

Some of the most common networking protocols and standards include:

- Transmission Control Protocol (TCP) and Internet Protocol (IP): These are the two most important protocols in the TCP/IP suite, which is the foundation of the internet. TCP provides reliable data transmission, while IP provides routing and addressing.
 - IP - It is responsible for routing and forwarding data packets from one device to another, based on their unique addressing scheme.
 - TCP - guarantees that all information packets arrive at their intended destination in the correct order and without errors.
- User Datagram Protocol (UDP): UDP is a simpler protocol than TCP that does not provide guaranteed delivery of data. It is often used for applications that do not require reliable data transmission, such as streaming media.
 - connectionless protocol that operates on top of the Internet Protocol (IP)
 - It is faster than Transmission Control Protocol (TCP) because it does not guarantee the delivery of packets or in order delivery, making it less reliable.
 - commonly used for time-sensitive applications such as online gaming, video streaming, and voice-over-IP (VoIP), where speed and efficiency are more important than reliability.
 - does not provide flow control or error correction like TCP

network standards

- File Transfer Protocol (FTP): FTP is a protocol for transferring files between two devices on a network. It is a reliable protocol that ensures that the files are transferred correctly.
 - It provides a simple way of sending and receiving files, making it an essential tool for businesses that need to transfer large amounts of data.
- Simple Mail Transfer Protocol (SMTP): SMTP is a protocol for sending email messages. It is a reliable protocol that ensures that the email messages are delivered to the correct recipient.
- Hypertext Transfer Protocol (HTTP): HTTP is a protocol for transferring web pages over the internet. It is a stateless protocol that does not track the state of a web session.
- It defines how web clients such as browsers communicate with servers to retrieve information.

These are just a few of the many networking protocols and standards that are in use today. These protocols and standards allow different devices to communicate with each other and make the internet possible.

Here are some additional benefits of using networking protocols and standards:

- Efficiency: Protocols and standards help to ensure that data is transmitted efficiently over a network. This is because they define the exact format of the data, which allows the devices to process it quickly and easily.
- Security: Protocols and standards can help to improve the security of a network. This is because they define the methods that are used to authenticate users and encrypt data.
- Interoperability: Protocols and standards help to ensure that different devices can interoperate with each other. This means that devices from different manufacturers can work together on the same network.
- Improved Network Performance – Adhering to standard formatting protocols helps to minimize data redundancy, optimize bandwidth usage and reduce network congestion, ultimately leading to better network performance.
- Facilitates Network Management – Protocols provide a standard set of rules for handling data transfer and adjustment of settings, making it easy for IT professionals to manage the connected devices on the network.

Overall, networking protocols and standards are essential for the efficient and secure operation of networks. They provide a common language that devices can use to communicate with each other, which makes the internet possible.

❖ Reference model and layered architectures in networking

A reference model is a conceptual framework that describes the different functions of a network. It is used to help network designers and engineers understand how networks work and to ensure that different parts of a network can work together.

A layered architecture is a way of organizing the different functions of a network into layers. Each layer performs a specific function, and the layers interact with each other to provide the overall functionality of the network.

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There are two main reference models for networking: the Open Systems Interconnection (OSI) model and the TCP/IP model.

- The OSI model is a seven-layer model that describes the different functions of a network. The layers are:
 - Physical Layer – Its function is to transmit individual bits from one node to another over a physical medium.
 - Data Link Layer – It is responsible for the reliable transfer of data frames from one node to another connected by the physical layer.
 - Network Layer – It manages the delivery of individual data packets from source to destination through appropriate addressing and routing.
 - Transport Layer – It is responsible for delivery of the entire message from the source host to destination host.
 - Session Layer – It establishes sessions between users and offers services like dialog control and synchronization.
 - Presentation Layer – It monitors syntax and semantics of transmitted information through translation, compression, and encryption.
 - Application Layer – It provides high-level APIs (application program interface) to the users.
- The TCP/IP model is a four-layer model that describes the different functions of a network. The layers are:
 - Network access layer
 - Internet layer
 - Transport layer
 - Application layer

Both the OSI model and the TCP/IP model are used to help network designers and engineers understand how networks work. However, the OSI model is more detailed than the TCP/IP model, and it is not as widely used in practice.

Layered architectures are often used in networking because they help to simplify the design and implementation of networks. By dividing the different functions of a network into layers, it is easier to understand how the network works and to troubleshoot problems.

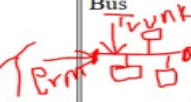
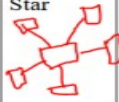
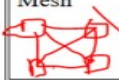
Here are some of the benefits of using layered architectures in networking:

- Simplicity: Layered architectures make it easier to understand how networks work. This is because the different functions of a network are organized into layers, and each layer performs a specific function.
- Scalability: Layered architectures can be scaled to accommodate different network sizes and requirements. This is because the different layers can be implemented independently of each other.
- Fault tolerance: Layered architectures can be designed to be fault-tolerant. This means that if one layer fails, the other layers can continue to operate.

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Overall, layered architectures are a valuable tool for network designers and engineers. They help to simplify the design and implementation of networks, and they can make networks more scalable and fault-tolerant.

❖ Network topology, media type and network component

Topology	Advantages	Disadvantages
Bus 	- Use of cable is economical. - Media is inexpensive and easy to work with. - System is simple and reliable. - Bus is easy to extend.	- Network can slow down in heavy traffic. - Problems are difficult to isolate. - Cable break can affect many users.
Ring 	- System provides equal access for all computers. - Performance is even despite many users.	- Failure of one computer can affect the rest of the network. - Problems are hard to isolate. - Network reconfiguration disrupts operation.
Star 	- Modifying system and adding new computers is easy. - Failure of one computer does not affect the rest of the network.	- If the centralized point fails, the network fails.
Mesh 	- System provides increased redundancy and reliability as well as ease of troubleshooting.	- System is expensive to install because it uses a lot of cabling.

$L = \frac{N(N-1)}{2}$

Network topologies, media types, and network components are all important concepts in networking.

Network topologies describe the physical layout of a network. They can be classified into two main types: physical topologies and logical topologies.

- Physical topologies describe the actual physical layout of the network, such as the way that the cables are connected.
- Logical topologies describe the way that data is transmitted over the network, regardless of the physical layout.

Some of the most common network topologies include:

- Point-to-point topology: A point-to-point topology is a simple network topology that connects two devices directly.
- Bus topology: A bus topology is a network topology in which all devices are connected to a single cable.
- Star topology: A star topology is a network topology in which all devices are connected to a central hub.
- Ring topology: A ring topology is a network topology in which all devices are connected in a loop.
- Mesh topology: A mesh topology is a network topology in which all devices are connected to each other.

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Media types describe the physical medium that is used to transmit data over a network. Some of the most common media types include:

- Twisted-pair cable: Twisted-pair cable is a type of copper cable that is commonly used for Ethernet networks.
- Coaxial cable: Coaxial cable is a type of copper cable that is commonly used for cable television networks.
- Fiber optic cable: Fiber optic cable is a type of cable that uses light to transmit data.

Network components are the physical devices that make up a network. Some of the most common network components include:

- Routers: Routers are devices that route data between different networks.
- Switches: Switches are devices that connect devices on the same network.
- Hubs: Hubs are devices that connect devices on the same network, but they do not provide any routing functionality.
- Servers: Servers are devices that provide shared resources to other devices on the network.
- Workstations: Workstations are devices that are used by users to access the network.

1. Linear Bus Topology – main run cable(node) are connected linear cable
2. Star Topology – each node connected directly to the central network called hub/switch
3. Ring Topology – connect computer on a single circle of cable
4. Mesh Topology - each computer is connected to every other computer by separate cabling
5. Hybrid - By modifying and combining like stat bus, star ring
6. Tree topology - combines characteristics of linear bus and star topologies.

❖ Media type

These transmission media may be of two types –

- **Guided** – In guided media, transmitted data travels through cabling system that has a fixed path. For example, copper wires, fibre optic wires, etc.
 - **Unguided** – In unguided media, transmitted data travels through free space in form of electromagnetic signal. For example, radio waves, lasers, etc.
1. Twisted Pair Cable
 - used for transmitting signals because of good performance at low costs.
 - are most commonly used in telephone lines.
 - However, if two or more wires are lying together, they can interfere with each other's signals. To reduce this electromagnetic interference, pair of copper wires are twisted together in helical shape like a DNA molecule. Such twisted copper wires are called **twisted pair**. To reduce interference between nearby twisted pairs, the twist rates are different for each pair.

Advantages of twisted pair cable

Twisted pair cable are the oldest and most popular cables all over the world. This is due to the many advantages that they offer –

- Trained personnel easily available due to shallow learning curve

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- Can be used for both analog and digital transmissions
- Least expensive for short distances
- Entire network does not go down if a part of network is damaged

Disadvantages of twisted pair cable

With its many advantages, twisted pair cables offer some disadvantages too –

- Signal cannot travel long distances without repeaters
- High error rate for distances greater than 100m
- Very thin and hence breaks easily
- Not suitable for broadband connections

2. Shielding twisted pair cable F

3. Coaxial cable

- Copper cables with better **shielding** than twisted pair cables, so that transmitted signals may travel longer distances at higher speeds. A coaxial cable consists of these layers, starting from the innermost –
- Stiff copper wire as **core**
- **Insulating material** surrounding the core
- Closely woven braided mesh of **conducting material** surrounding the **insulator**
- Protective **plastic sheath** encasing the wire

Coaxial cables are widely used for **cable TV** connections and **LANs**.

4. Optical Fibre

- Thin glass or plastic threads used to transmit data using light waves

5. Infrared

- Low frequency infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc. Infrared signals can propagate within a room but cannot penetrate walls. However, due to such short range, it is considered to be one of the most secure transmission modes.

6. Radio wave

- Transmission of data using radio frequencies is called **radio-wave transmission**. We all are familiar with radio channels that broadcast entertainment programs. Radio stations transmit radio waves using **transmitters**, which are received by the receiver installed in our devices.
- Both transmitters and receivers use antennas to radiate or capture radio signals. These radio frequencies can also be used for **direct voice communication** within the **allocated range**. This range is usually 10 miles.

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❖ Network component

Computer network components are the *major parts* which are needed to *install the software*. Some important network components are **NIC**, **switch**, **cable**, **hub**, **router**, and **modem**. Depending

on the type of network that we need to install, some network components can also be removed. For example, the wireless network does not require a cable.

1. NIC

- NIC stands for network interface card.
- NIC is a hardware component used to connect a computer with another computer onto a network
- It can support a transfer rate of 10,100 to 1000 Mb/s.
- The MAC address or physical address is encoded on the network card chip which is assigned by the IEEE to identify a network card uniquely. The MAC address is stored in the PROM (Programmable read-only memory).

2. Hub

- A Hub is a hardware device that divides the network connection among multiple devices. When computer requests for some information from a network, it first sends the request to the Hub through cable. Hub will broadcast this request to the entire network. All the devices will check whether the request belongs to them or not. If not, the request will be dropped.
- The process used by the Hub consumes more bandwidth and limits the amount of communication. Nowadays, the use of hub is obsolete, and it is replaced by more advanced computer network components such as Switches, Routers.

3. Switch

- A switch is a hardware device that connects multiple devices on a computer network. A Switch contains more advanced features than Hub. The Switch contains the updated table that decides where the data is transmitted or not. Switch delivers the message to the correct destination based on the physical address present in the incoming message. A Switch does not broadcast the message to the entire network like the Hub. It determines the device to whom the message is to be transmitted. Therefore, we can say that switch provides a direct connection between the source and destination. It increases the speed of the network.

4. Router

- A router is a hardware device which is used to connect a LAN with an internet connection. It is used to receive, analyze and forward the incoming packets to another network.
- A router works in a **Layer 3 (Network layer)** of the OSI Reference model.
- A router forwards the packet based on the information available in the routing table.
- It determines the best path from the available paths for the transmission of the packet.

5. Modem

- A modem is a hardware device that allows the computer to connect to the internet over the existing telephone line.
- A modem is not integrated with the motherboard rather than it is installed on the PCI slot found on the motherboard.
- It stands for Modulator/Demodulator. It converts the digital data into an analog signal over the telephone lines.

Hardware – device and media

Software – service and process

❖ OSI layer at various network device operate

Here is a table of the OSI layers at which various network devices operate:

Network Device	OSI Layer
Routers	Network layer
Switches	Data link layer
Hubs	Physical layer
Servers	Application layer
Workstations	Application layer

As you can see, routers operate at the network layer, which is responsible for routing data between different networks. Switches operate at the data link layer, which is responsible for connecting devices on the same network. Hubs operate at the physical layer, which is responsible for transmitting data over the physical medium. Servers and workstations operate at the application layer, which is responsible for providing services to users.

It is important to note that these are just general guidelines. The specific OSI layers at which a device operates may vary depending on the device's capabilities and the network architecture.

Here are some additional details about the OSI layers at which various network devices operate:

- Routers operate at the network layer by using routing protocols to determine the best path for data to travel between different networks.
- Switches operate at the data link layer by using Media Access Control (MAC) addresses to connect devices on the same network.
- Hubs operate at the physical layer by simply repeating data that is received on one port to all other ports.
- Servers operate at the application layer by providing services to users, such as file sharing, web hosting, and email.
- Workstations operate at the application layer by allowing users to access network resources, such as files, printers, and applications.

Two Sets of Layers

A. Application

1. Session
2. Presentation
3. Application

B. Data Transport

1. Physical
2. Data Link
3. Network
4. Transport

❖ **Apply basic cabling and network design technique to connect device**

Here are some basic cabling and network design techniques that can be used to connect devices:

- Choose the right type of cable: The type of cable that you choose will depend on the distance between the devices, the amount of data that needs to be transferred, and the environment in which the cable will be installed.
- Use the correct connectors: The connectors that you use will also depend on the type of cable that you are using. Make sure that the connectors are compatible with the devices that you are connecting.
- Install the cables correctly: When installing cables, it is important to follow the manufacturer's instructions. This will help to ensure that the cables are installed correctly and that they will not be damaged.
- Label the cables: Labeling the cables will help you to identify them and to troubleshoot problems.
- Plan the network topology: The network topology is the physical layout of the network. Planning the network topology will help you to choose the right type of cable and to install the cables correctly.
- Use a cable tester: A cable tester can be used to test the cables and to make sure that they are installed correctly.

Here are some additional tips for connecting devices:

- Use a cable guide: A cable guide can help to keep the cables organized and to prevent them from being damaged.
- Secure the cables: The cables should be secured to prevent them from being pulled or damaged.
- Test the connections: Once the cables are installed, you should test the connections to make sure that they are working properly.